



Product Requirements Document (PRD)

Product Vision

A decentralized identity and reputation tracker that empowers users to build trust across marketplaces and communities without relying on centralized intermediaries.

Goals

- Enable users to register and verify their identity via Bitcoin addresses.
- Provide transparent, immutable reputation scores.
- Allow third-party platforms to integrate reputation data.

Functional Requirements

1. **Identity Management**
 - Register Bitcoin address.
 - Link optional metadata (username, profile info).
 - Verify ownership via cryptographic signature.
2. **Reputation System**
 - Record ratings/reviews tied to transactions.
 - Calculate scores using weighted algorithms.
 - Display reputation history on user dashboard.
3. **Privacy & Security**
 - Zero-knowledge proofs for selective disclosure.
 - Encryption of sensitive metadata.
 - Dispute resolution via smart contracts.
4. **Integration APIs**
 - REST/GraphQL endpoints for external platforms.
 - Reputation query by Bitcoin address.
 - Event subscription for updates.
5. **Frontend Web Application**
 - User-friendly dashboard.
 - Reputation visualization (graphs, badges).
 - Identity management tools.

Non-Functional Requirements

- **Scalability:** Handle thousands of reputation events per second.
- **Security:** Smart contract audits, cryptographic verification.
- **Usability:** Simple onboarding for non-technical users.
- **Interoperability:** Compatible with marketplaces, DAOs, and social apps.